

EE3621: Digital Signal Processing

Lecture 21: Linear phase FIR filter, characteristic response, location of zeros (III B.Tech EEE)

September 23, 2026

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Derive** linear phase conditions for symmetric and antisymmetric FIR impulse responses.
 2. **Classify** FIR filters into Types I, II, III, and IV based on length N and symmetry.
 3. **Determine** boundary zero constraints at $\omega = 0$ and $\omega = \pi$ for each FIR type.
 4. **Analyze** zero locations in the complex z -plane.
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2. MATHEMATICAL FOUNDATIONS

2.1 The 4 Types of Linear-Phase FIR Filters

— Type — Symmetry — Length N (Order $M = N - 1$) — Amplitude Function $A(\omega)$ — Mandatory Boundary Zeros — — :—: — :—: — :—: — :—: — :—: — — **Type I** — Symmetric ($h[n] = h[N - 1 - n]$) — N Odd (M Even) — $\sum a[k] \cos(\omega k)$ — None — — **Type II** — Symmetric ($h[n] = h[N - 1 - n]$) — N Even (M Odd) — $\sum b[k] \cos(\omega(k - 1/2))$ — Zero at $\omega = \pi$ ($z = -1$) — — **Type III** — Antisymmetric ($h[n] = -h[N - 1 - n]$) — N Odd (M Even) — $\sum c[k] \sin(\omega k)$ — Zeros at $\omega = 0$ ($z = 1$) & $\omega = \pi$ ($z = -1$) — — **Type IV** — Antisymmetric ($h[n] = -h[N - 1 - n]$) — N Even (M Odd) — $\sum d[k] \sin(\omega(k - 1/2))$ — Zero at $\omega = 0$ ($z = 1$) —

2.2 Proof of Mandatory Zeros

- **Type II at $\omega = \pi$ ($z = -1$):**

$$H(-1) = \sum_{n=0}^{N-1} h[n](-1)^{-n} = \sum_{n=0}^{N/2-1} h[n][(-1)^{-n} + (-1)^{-(N-1-n)}]$$

Since N is even, $N - 1$ is odd $\implies (-1)^{-(N-1-n)} = -(-1)^{-n} \implies H(-1) = 0$. Therefore, a Type II filter always has a zero at $\omega = \pi$ and can **never** be used as a Highpass or Bandstop filter.

2.3 Zero Quadruplet Locations

Because $h[n]$ is real and symmetric/antisymmetric:

$$H(z) = \pm z^{-(N-1)} H(z^{-1})$$

If $z_0 = re^{j\theta}$ is a zero, then:

1. $z_0^* = re^{-j\theta}$ is a zero (conjugate symmetry for real coefficients).
2. $1/z_0 = \frac{1}{r}e^{-j\theta}$ is a zero (linear phase symmetry).
3. $1/z_0^* = \frac{1}{r}e^{j\theta}$ is a zero.

Zeros off the unit circle always occur in groups of 4 (quadruplets).

Pictorial Representations

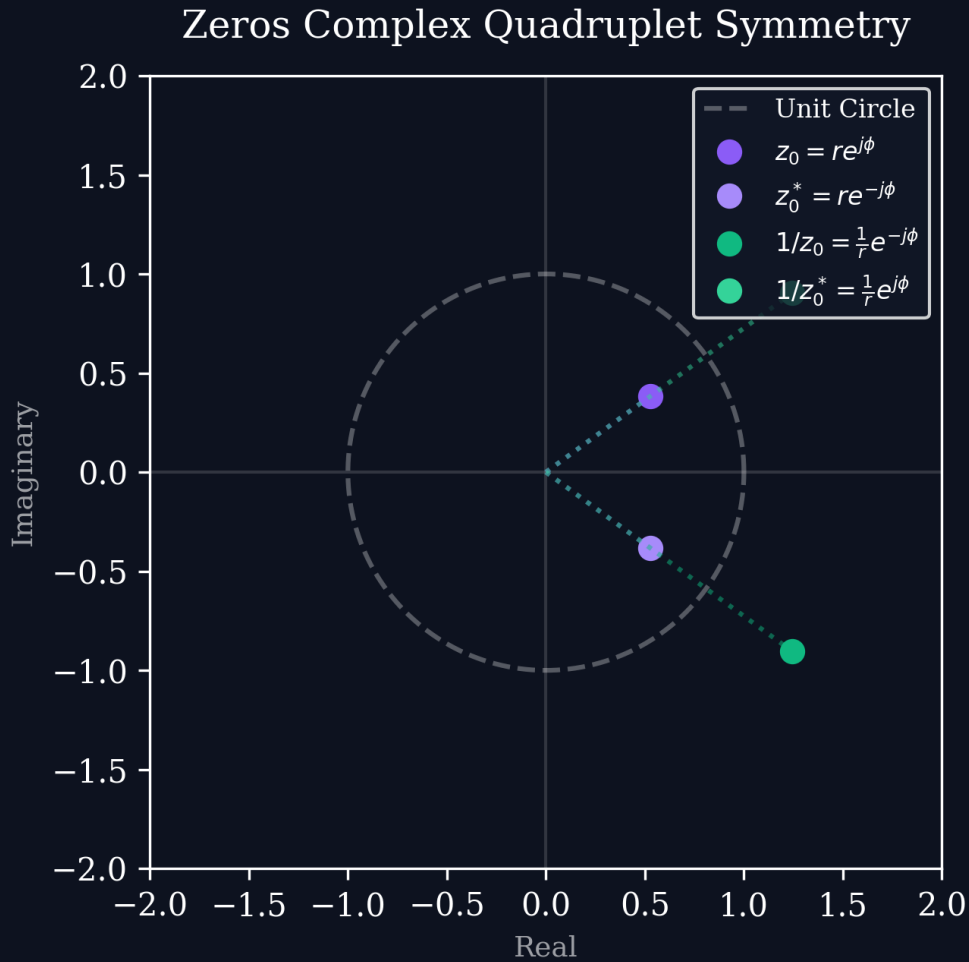


Figure 1: Zero Quadruplet

Frequency Constraints of Linear-Phase FIR Types

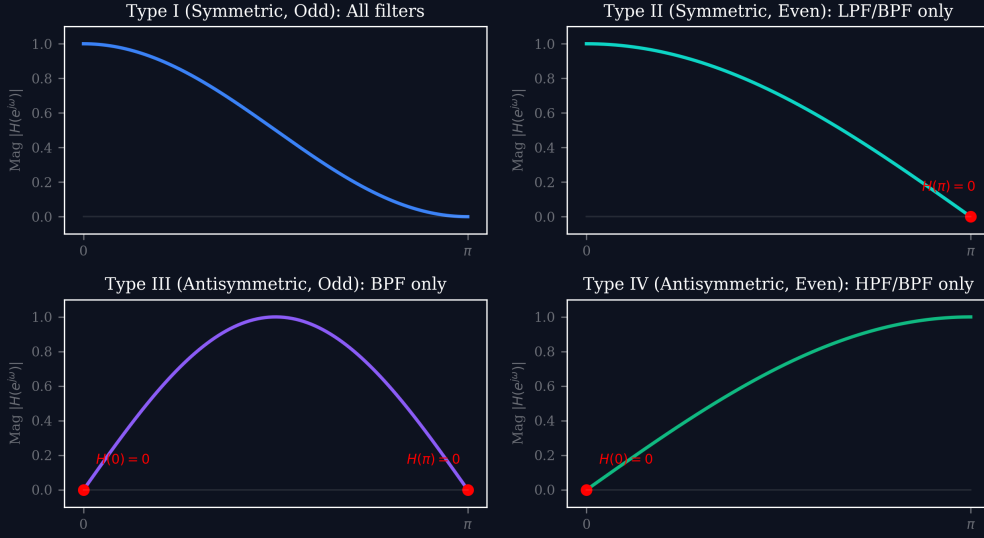


Figure 2: Fir Types

3. WORKED NUMERICAL EXAMPLES

Example 21.1: FIR Type Identification & Zero Locations

Problem: An FIR filter has transfer function $H(z) = 1 + 2z^{-1} + 3z^{-2} + 2z^{-3} + z^{-4}$. (a) Determine its FIR Type, filter length N , and group delay τ_g . (b) Can this filter be used as a Highpass filter?

Solution: (a) Coefficients: $h = [1, 2, 3, 2, 1]$.

- Length $N = 5$ (Odd), Order $M = 4$ (Even).
- Symmetry: $h[0] = h[4] = 1$, $h[1] = h[3] = 2$, $h[2] = 3 \implies$ **Symmetric**.
- Therefore, it is a **Type I FIR Filter**.
- Group delay: $\tau_g = \frac{N-1}{2} = \frac{5-1}{2} = 2$ samples.

(b) Highpass Feasibility: Check $H(e^{j\pi}) = H(-1) = 1 - 2 + 3 - 2 + 1 = 1 \neq 0$. Since Type I has no mandatory zeros at $\omega = \pi$, it can theoretically operate as a Highpass filter (though this specific impulse response has $H(e^{j0}) = 9$ and $H(e^{j\pi}) = 1$, making it a Lowpass filter).